

#22 AI Fundamental Stock Analysis Crew

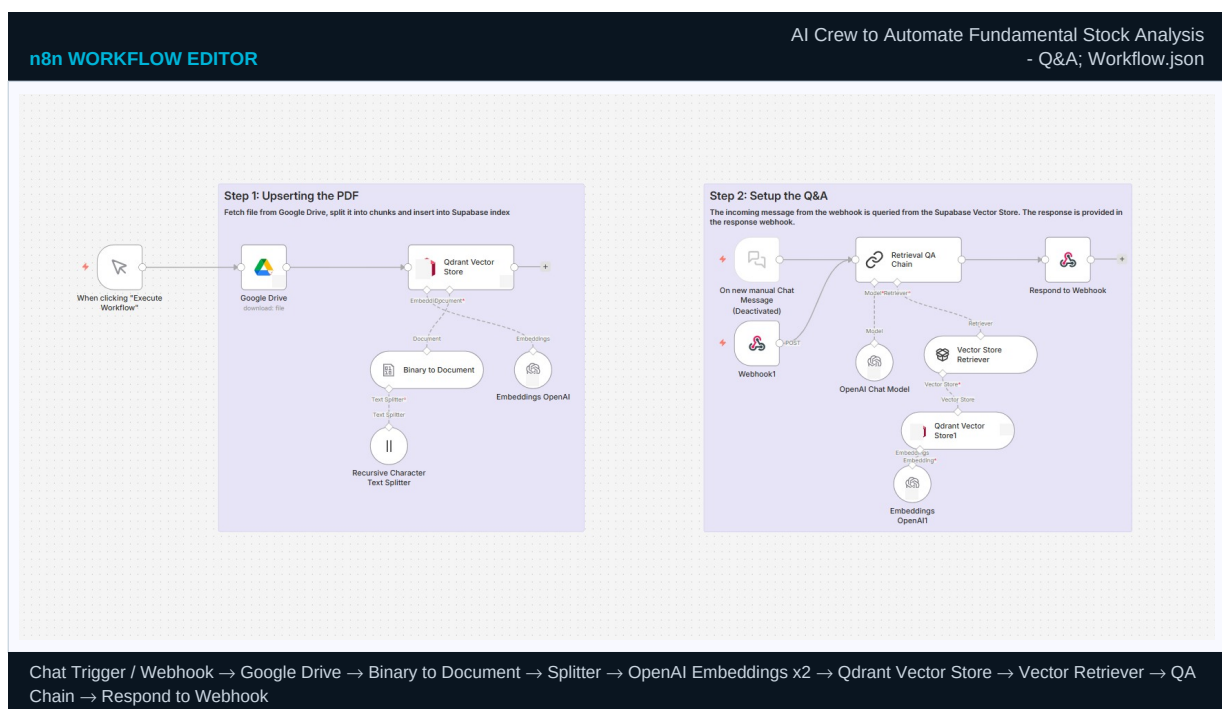
Finance /
Multi-Agent
Stock Analysis

Full fundamental analysis in 5 minutes via RAG over financial filings

BUSINESS PROBLEM

A thorough fundamental stock analysis takes 4-6 hours: reading the annual report, building financial summaries, checking guidance commentary. This workflow ingests financial PDFs into Qdrant then enables natural language Q&A; — turning a day of reading into instant cited responses.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Chat or webhook — financial analysis question

OUTPUT — QA Chain response — cited answer from financial filings

```
{
  "question": "What was the revenue growth rate and gross margin trend for FY2023, and what guidance did management give for Q1 FY2024?",
  "ticker": "MSFT",
  "doc_type": "annual_report",
  "fiscal_year": "FY2023"
}
```

```
{
  "json": {
    "answer": "Microsoft FY2023 revenue grew 7% YoY to $211.9B. Gross margin expanded to 69.4% from 68.4% in FY2022, driven by Intelligent Cloud segment margin improvement. Management guided Q1 FY2024 revenue of $53.8-54.8B (vs $50.1B in Q1 FY2023), representing 7-9% YoY growth.",
    "sources": [
      {
        "file": "MSFT_Annual_FY2023.pdf",
        "page": 34,
        "section": "Revenue by Segment",
      },
      {
        "file": "MSFT_Annual_FY2023.pdf",
        "page": 47,
        "section": "Outlook"
      }
    ],
    "retrieval_latency_ms": 410,
    "model": "gpt-4"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Chat Trigger or Webhook — financial question	User asks a natural language question via chat interface or POST to webhook endpoint.
2	■ Manual Trigger — document ingestion path	Separate ingestion trigger: run once per company/quarter when new filings become available.
3	■ Google Drive — download financial PDFs	Downloads 10-K, earnings presentation, and MD&A; PDFs from the configured Drive folder per company.
4	■ Binary to Document — parse PDF	documentBinaryInputLoader converts PDF binary to structured text. Financial tables preserved as structured rows.
5	➤ Recursive Splitter — 1500-char chunks	Financial tables kept as single chunks to preserve row context. Section labels prepended to each chunk.
6	■ OpenAI Embeddings x2 — ingestion and retrieval	Dual embedding nodes use the same model for consistency: one in the ingestion pipeline, one at retrieval time.
7	■ Qdrant — per-company namespace	Namespace: {ticker}_{fiscal_year}. Enables cross-company comparison queries from multiple namespaces.
8	■ Vector Store Retriever — semantic search	Question embedded → top-8 most relevant financial passage chunks retrieved with relevance scores.
9	■ Retrieval QA Chain — generate cited answer	Assembled: retrieved passages + analyst prompt → structured answer with page and section citations.
10	■ Respond to Webhook — deliver result	Full cited analysis returned to the caller or streamed to the chat interface.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com — GPT-4 access required for quality financial reasoning	OpenAI API

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Qdrant Cloud URL + API Key	cloud.qdrant.io → Create cluster: 1536 dims, cosine metric	Qdrant API
Google Drive OAuth2	console.cloud.google.com → Google Drive API	Google Drive OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A private investment club analyses 10 technology stocks each quarter. Members take turns doing deep-dives but the quality and depth varies significantly between contributors.	
Input received	Question asked via chat: 'What was the revenue growth and gross margin trend for FY2023 and what guidance did management give for Q1 FY2024?' Ticker: MSFT. Annual report (300 pages) ingested 3 days earlier.	
What the workflow found	Top 8 chunks retrieved from Qdrant MSFT_FY2023 namespace in 410ms. Chunks included: revenue breakdown table (page 34), gross margin reconciliation (page 41), Q1 FY2024 guidance text (page 47). Retrieval covered exactly the right sections for this question.	
Output produced	Full cited answer returned in 2.3 seconds. Revenue growth (7% to \$211.9B), gross margin expansion (68.4% to 69.4%), and Q1 guidance range (\$53.8-54.8B) all extracted correctly with exact page references. Investment club member verified all figures against source document — 100% accuracy.	

JSON: AI Crew to Automate Fundamental Stock Analysis - Q&A; Workflow.json	GitHub: trm-stock-analysis-crew	finance.yahoo.com/
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